

1 Optimization Results Tables

Table 1: Final optimization improvement over standard LwF for ImageNet $\rightarrow$ Scenes

Method	ImageNet Old Accuracy	Scenes New Accuracy	Old Task Improvement	New Task Improvement
Standard LwF	59.82%	56.27%	–	–
Optimized LwF	60.72%	58.06%	+0.90%	+1.79%

Table 2: Comparison between reproduced methods and the final optimized LwF method

Method	ImageNet Old Accuracy	Scenes New Accuracy	Main Observation
Fine-Tuning	48.69%	50.82%	Significant forgetting of the old task
Feature Extraction	62.89%	55.00%	Strong old-task preservation, weaker new-task adaptation
Standard LwF	59.82%	56.27%	Balanced continual learning baseline
Optimized LwF	60.72%	58.06%	Best new-task accuracy with strong old-task preservation

Table 3: Summary of the final optimization effect

Optimization Aspect	Effect
Scaled cosine-normalized classifier	Improved new-task decision boundary
Learnable scale parameter $\eta_0 = 30$	Sharper and more effective softmax logits
Boosted new-head learning rate, $LR_{newhead} = 5 \times LR_{base}$	Faster adaptation to MIT Indoor Scenes
LwF distillation loss preserved	Maintained old ImageNet knowledge
Final result	Improved Scenes accuracy without reducing ImageNet accuracy